

requiring exceptional power density and a high peak power rating, specifically in server and cloud environments. The converters are designed to operate with an input range of 40-60V, making them suitable for 48/54V applications commonly found in data centres and high-performance



computing environments. The module ensures a fully regulated output of 12.2V nominal without isolation, ensuring reliable performance. With efficiency levels reaching over 98%, the converters offer remarkable energy efficiency, even under typical conditions with a 54V input and half load. The module incorporates a comprehensive thermal management system to enable secure peak power operation, featuring fast and timed over-temperature protection. It also includes an alert signal for imminent thermal shutdown indication. The converters come equipped with various protection mechanisms, including input under-voltage lock-out and output short-circuit, over-voltage, and over-current protection. The output voltage of the converters can be adjusted within the range of 8V to 13.2V, and they offer hardware pins for remote sensing, power good indication, and remote control. With a power output of 1,600W (2,320W peak) and an output current of 136A (200A peak), the BMR351 series from Flex Power Modules provides high performance and power density for demanding server and cloud applications.

Flex Power Modules
<https://flexpowermodules.com>

SSD for gaming

Consistent Infosystems, an Indian brand, has unveiled their NVMe 4th Gen solid-state drive (SSD) offering exceptional speed and performance. This NVMe SSD utilises the PCIe Gen4 x4 interface and boasts speeds up to 10 times faster than traditional SSDs.

With a read speed of up to 7,300MB per second, the SSD enables faster application loading, quicker boot times, and an improved overall user experience. The SSD incorporates advanced



features such as 3D NAND technology, which enhances speed, storage capacity, efficiency, and energy consumption. Additionally, it includes SLC caching for high transfer speeds, even on flash memory that typically has lower write speeds, and LDPC Advanced for reliable transmission over noisy channels. Consistent Infosystems aims to provide customers with an exceptional user experience by leveraging the latest technologies in their products. The NVMe 4th Gen SSD is available in three storage capacity variants: 512GB, 1TB, and 2TB, with read speeds of 5,000Mbps, 7,000Mbps, and 7,300Mbps, respectively.

Consistent Infosystems
<https://consistent.in>

EMBEDDED

Camera module for AI application

Innodisk has introduced a highly customisable camera modules designed for artificial intelligence (AI) and computer vision applications. These cam-



era modules offer several advantages, including customised development, platform compatibility, and internet service provider (ISP) adaptation. The modules seamlessly integrate with major operating systems and software, simplifying the integration process. With ISP technology, the modules can be easily adjusted to optimise performance based on each customer's unique application. The camera modules find applications in various scenarios such as AI image recogni-

tion, where compact mobile industry processor interface (MIPI) cameras are integrated into systems like parking meters and production lines for automatic detection. They provide high resolution solutions up to 1920 × 1080 pixels. The camera modules are expected to make significant contributions to the field of AI computer vision applications worldwide.

Innodisk
<https://www.innodisk.com>

RT Ethernet PHY devices

Microchip Technology Inc. has introduced radiation-tolerant (RT) Ethernet PHY devices designed for space applications. These devices support the Serial Gigabit Media Independent



Interface (SGMII) and Quad Serial Gigabit Media-Independent

Interface (QSGMII), offering increased versatility by supporting both copper and fibre interfaces. By reducing the number of signal pins required, the devices free up space in the host device. The low-power quad-port Gigabit Ethernet transceiver is specifically designed for the aerospace environment and is compatible with copper and fibre interfaces. It incorporates a quad port to support 10, 100, and 1000BASE-T Ethernet connections, ensuring optimal speed and reach. With its ability to withstand extreme temperatures, vacuum conditions, and enhanced radiation performance, the device provides reliable connectivity in harsh space environments.

Microchip Technology
<https://www.microchip.com>

Solderable SoM

iWave has introduced a solderable system on module (SoM) featuring the i.MX 9 applications processor. The compact SoM offers a durable form factor and a wide range of interfaces for versatile connectivity, making it suitable for various applications